

A Retrospective Analysis of Internet Crimes Against Children Across U.S. Task Forces, 2010–2026

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Abstract

The history of the internet has been written largely from the perspective of platforms, entrepreneurs, and policy makers. Largely absent from that record is a systematic account of how the same infrastructure transformed the landscape of child exploitation—and how law enforcement institutions adapted, strained, and responded across three decades of technological change. This gap is not incidental: internet-facilitated crimes against children have scaled alongside the web itself, yet their history remains fragmented across jurisdictions, buried in press releases, and invisible at the institutional level.

This paper presents the first longitudinal case study analysis of the Internet Crimes Against Children (ICAC) Task Force Program across 2010–2026, organized across four technological eras: the social platform era (2010–2014), the mobile-first transition (2015–2018), platform proliferation and expansion (2019–2022), and the social media and generative AI period (2023–2026). Drawing on 530 publicly available case reports processed through CaseLinker—an open-source retrospective case analysis system—we develop 100 structured case studies that trace how perpetrator methodologies, platform surfaces, and law enforcement responses co-evolved across each era.

The contribution of this paper is both historical and methodological. Historically, the case studies span platform involvement, victim and perpetrator demographics, investigation type (proactive, undercover, and reactive), geographic and jurisdictional patterns, and agency coordination networks. Each era is characterized by shifting perpetrator methodologies, evolving platform surfaces, and corresponding law enforcement adaptations—patterns that are invisible at the individual case level and only emerge through systematic cross-case analysis. Methodologically, CaseLinker provides an auditable, open-source infrastructure for retrospective case analysis, designed to be replicable and extensible by other researchers working in sensitive domains. We situate this work within the broader Trust and Safety literature, treating ICAC task forces as an example of institutional response to digital harm—

one operating under escalating volume, with 20.5 million CyberTip reports filed in 2024 alone—and whose operational history offers grounded insight into how emerging technologies have compounded harm against vulnerable populations.

Research Methodology

This research is grounded in CaseLinker, an open-source retrospective case analysis system developed to process, structure, and surface patterns from publicly available ICAC press releases and task force case reports. The system is designed to be auditable, interpretable, and appropriate for the sensitivity of the subject matter—operating exclusively on already-public, already-redacted case summaries and producing no outputs that identify individuals.

Feature extraction. Extraction operates in two layers. The primary layer is a deterministic pattern extraction: domain calibrated set of regular expressions and linguistic rules that extract structured fields from case text—offense type, platform mentioned, victim age range, perpetrator–victim relationship, investigation classification (proactive, online, undercover, vs. reactive), prosecutorial outcome, and agency involvement, among others. Every extracted feature is traceable to its source text in the database, enabling full auditability. The secondary layer is a Named Entity Recognition system (Stanford Stanza) that augments pattern extraction with semantic entity detection—particularly for law enforcement organization identification, where surface-form variation across agencies makes rule-based approaches brittle. NER outputs are merged with pattern outputs under a precedence system that preserves deterministic primacy; the machine learning layer is additive and optional, not load-bearing.

Analysis stack. Case-level features feed a multi-component analysis stack. A 9 dimension facet tree supports cohort-level navigation and constraint-based case filtering—enabling structured queries such as grooming-platform cases involving victims under 12 in reactive investigations from 2018–2022, returned as a structured case cohort with membership counts and facet paths. A Jaccard-similarity clustering layer groups cases by feature overlap, surfacing recurring case profiles that cut across jurisdictions and years. A rule-based triage system with documented, inspectable scoring weights assigns priority tiers to cases based on severity indicators, and a Random Forest classifier trained on triage labels supports further analysis—with outputs explicitly downstream of and accountable to the rule layer.

Case study selection. The 100 case studies will be selected using stratified sampling across the four historical eras and five analytic dimensions—platform, demographics, investigation type, geography, and agency network—to ensure coverage of both dominant patterns and structurally important edge cases. Each case study is written as a structured narrative: platform context, perpetrator methodology, investigative approach, prosecutorial outcome,

and relationship to the broader technological moment of its era. The goal is not statistical generalization from the sample, but grounded historical illustration of patterns that aggregate analysis surfaces and that individual case reading cannot reveal. All case data is drawn from public sources. The full source corpus, extraction code, and analysis pipeline are released as open-source software to support replication and extension by other researchers.

CaseLinker currently contains 530 cases, 5516 extracted features, and 6 sources. The analysis pipeline is operational and will be used to guide case study selection and cross-era pattern analysis throughout the research. This project received a determination from the University of Massachusetts Amherst Human Research Protection Office (HRPO Determination #7668); this research does not contain private or identifiable information under federal regulations [45 CFR 46.102(f)(1), (2)]

Initial References

- [1] National Center for Missing & Exploited Children. (2024). *2024 CyberTipline Data Report*. <https://www.missingkids.org/gethelpnow/cybertipline/cybertiplinedata>
- [2] Office of Juvenile Justice and Delinquency Prevention. (2024). *Internet Crimes Against Children Task Force Program*. <https://ojjdp.ojp.gov/programs/internet-crimes-against-children>
- [3] Souras, Y. (2025, March 26). *Testimony before the U.S. House Committee on Energy and Commerce regarding CyberTipline reporting and the REPORT Act*. <https://www.congress.gov/119/meeting/house/118066/witnesses/HHRG-119-IF17-Wstate-SourasY-20250326.pdf>
- [4] Internet Watch Foundation. (2024, July). *How AI is being abused to create child sexual abuse imagery: Updated report*. <https://www.iwf.org.uk/about-us/why-we-exist/our-research/how-ai-is-being-abused-to-create-child-sexual-abuse-imagery/>
- [5] eSafety Commissioner. (2026, January 8). *Generative AI and child safety: A convergence of innovation and exploitation*. <https://www.esafety.gov.au/newsroom/blogs/generative-ai-and-child-safety-a-convergence-of-innovation-and-exploitation>
- [6] Technology Coalition. (2023). *An update on voluntary detection of CSAM*. <https://technologycoalition.org/resources/update-on-voluntary-detection-of-csam/>
- [7] Thorn. (2023, September). *Hashing and matching is core to proactive CSAM detection*. <https://safer.io/resources/hashing-and-matching-is-core-to-proactive-csam-detection/>
- [8] Thorn. (2025, January). *Cross-platform CSAM hash sharing for platforms*. <https://safer.io/resources/cross-platform-csam-hash-sharing-for-platforms/>

[//safer.io/resources/optimize-csam-detection-with-saferlist/](https://safer.io/resources/optimize-csam-detection-with-saferlist/)

- [9] Steinebach, M. (2023). An analysis of PhotoDNA. *ARES 2023* (18th International Conference on Availability, Reliability and Security). <https://dl.acm.org/doi/fullHtml/10.1145/3600160.3605048>
- [10] Deryck, M., Leblanc-Albarel, D., & Preneel, B. (2026). White-box attacks on PhotoDNA perceptual hash function. *IACR ePrint*, 2026/486. <https://eprint.iacr.org/2026/486.pdf>
- [11] United Nations Interregional Crime and Justice Research Institute. (2022). *AI for Safer Children Global Hub*. <https://unicri.org/topics/AI-for-Safer-Children>
- [12] Burns, C. M., Morley, J., Bradshaw, R., & Domene, J. (2008). The emotional impact on and coping strategies employed by police teams investigating internet child exploitation. *Traumatology*, 14(2), 20–31.
- [13] Perez, L. M., Jones, J., Englert, D. R., & Sachau, D. (2010). Secondary traumatic stress and burnout among law enforcement investigators exposed to disturbing media images. *Journal of Police and Criminal Psychology*, 25(2), 113–124.
- [14] Klein, G. (1998). *Sources of Power: How People Make Decisions*. MIT Press.
- [15] ICAC Task Force Public Case Reports — Arizona, Idaho, Texas, Georgia, and additional task forces (2010–2026). Collected and indexed via the CaseLinker public corpus.
- [16] Ramachandran, M. (2026). *CaseLinker: An open-source system for cross-case analysis of Internet Crimes Against Children reports*. Technical Report #1. University of Massachusetts Amherst. <https://doi.org/10.48550/arXiv.2603.18020>
- [17] Ramachandran, M. (2026, March). *CaseLinker: Interpretable ML approaches for analyzing Internet Crimes Against Children reports*. Technical Report #2. <https://mrinaalr.github.io/website/CaseLinker-%20Interpretable%20ML%20Approaches%20for%20Analyzing%20Internet%20Crimes%20Against%20Children%20Reports.pdf>